

Vol 3 Chapter 2 — Magic Numbers from Substrate Spin-Orbit

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2026-05-23 Saturday

Chapter 2 — Magic Numbers from Substrate Spin-Orbit

Nuclear magic numbers — the values of Z (proton count) or N (neutron count) at which nuclei are unusually stable — are 2, 8, 20, 28, 50, 82, 126. The Mayer–Jensen 1949 nuclear shell model explained them via a strong spin-orbit coupling that splits the $j = l + 1/2$ and $j = l - 1/2$ subshells. The coupling strength is fit to reproduce the seven observed magic numbers.

BST derives all seven exactly from a single substrate-primary ratio.

2.1 The substrate spin-orbit coupling

The substrate’s natural spin-orbit coupling is

$$\kappa_{ls} = \frac{C_2}{n_C} = \frac{6}{5}.$$

Inserting this single value into the standard Mayer–Jensen formalism reproduces all seven magic numbers — 2, 8, 20, 28, 50, 82, 126 — exactly. No additional tuning. 7/7 match at D-tier.

2.2 The derivation

The shell-model orbital sequence under harmonic oscillator + spin-orbit gives splittings that produce shell closures at specific cumulative occupation numbers. Standard physics chooses the spin-orbit strength to make the splittings produce the observed magic numbers. BST derives the strength from substrate primaries.

The substrate-mechanism reading: the spin-orbit coupling is the substrate’s natural Casimir-to-dimension ratio, with $C_2 = 6$ being the substrate Casimir eigenvalue (Volume 1 Chapter 5) and $n_C = 5$ being the substrate complex dimension (Volume 0 Chapter 2). The ratio 6/5 enters the shell-model Hamiltonian’s spin-orbit term directly.

2.3 Match precision

All seven magic numbers match exactly (integer agreement). No deviation. D-tier ratified.

2.4 What comes next

Chapter 3 — Nuclear Shell Model — extends to the top 30 isotopes’ shell-filling structure.

Where to look this up: Spin-orbit coupling derivation: T-numbered theorem in substrate-physics catalog. Mayer–Jensen 1949 shell model: original papers. Standard nuclear reference: Krane.